

Evidence Appendix: Supplement

MultiPlanetary Exchange System · Milestone 2 · Design v3

About this supplement

This supplement to the evidence appendix ([appendix_short.pdf](#)) holds the detail behind it: complete traces, every launch and every ledger change. Its tables are numbered B; tables numbered A are in the appendix. Every result comes from the same simulator runs as the appendix, and the machine-readable traces are in `milestone2/traces/`.

F. Message formats and sizes

Every message has a fixed-width binary layout. The tables below give every field, and the record sizes are the ones the simulator uses for packing. Integers are unsigned big-endian unless a field says otherwise. Times are signed 64-bit integer milliseconds of TDB since the epoch. Money is in integer cents, and share quantities are whole shares. Each record carries the signature of the principal that issued it, so it can be checked at any exchange whatever session or forwarder carried it.

Table B1: Common header of every application record.

Field	Bytes and meaning
Type	1: Record type code
Version	1: Format version (1)
Length	2: Record length in bytes
Record ID	12: 32-bit principal prefix and 64-bit counter; never reused
Created	8: Signed 64-bit integer milliseconds of TDB since the epoch
Signature	64: Ed25519 by the issuing principal over every other byte of the record
Total	88

Table B2: Fixed-width body of each record type. The totals are the sizes the simulator charges (`protocol.SIZE`); the build checks that the fields add up to them. A STATUS record grows by 16 bytes for each entry after the first.

Record	Bytes	Fields after the common header (bytes)
TRANSFER	176	Source exchange 1; Destination exchange 1; Purpose 1; Asset 1; Owner account 8; Amount (units or cents) 8; Number n to this destination 8; Reference ID (order, contract, request) 12; Beneficiary account 8; Terms digest (SHA-256) 32; Reserved 8
RECEIVED	120	Transfer ID 12; Source exchange 1; Result (credited, restricted) 1; Import time 8; Watermark W for this source 8; Reserved 2
ORDER	128	Account 8; Host 1; Asset 1; Side 1; Time in force 1; Quantity 8; Limit price (cents) 8; Expiry 8; Disposition (keep, home) 1; Reserved 3
CANCEL	128	Account 8; Target order ID 12; Reserved 20
WITHDRAW	128	Account 8; Asset 1; Destination exchange 1; Amount 8; Reserved 22
CLOSE	128	Account 8; Reserved 32
CINSTR	256	Account 8; Contract ID 12; Role (long, short) 1; Quantity Q 8; Multiplier 4; Opening price K 8; Floor 8; Cap 8; Five observation times 40; Maturity 8; Fixing time 8; Opening deadline 8; Counterparty account 8; Price source 8; Margin 8; Reserved 23
STATUS	128	Host exchange 1; Home exchange 1; Entry count m (1 to 52) 2; As-of time 8; Reserved 12; Entry 1: account 8, contract counter 6, state 1, flags 1 16
PRICE	200	Contract ID 12; Observation number 1; Index value (fixed point) 8; Observation time 8; Price source 8; Source attestation (Ed25519) 64; Reserved 11
MDS	152	Market exchange 1; Asset 1; Book version 8; Best bid (cents) 8; Bid size 4; Best ask (cents) 8; Ask size 4; Last trade price (cents) 8; Last trade size 4; Last trade time 8; Reserved 10
MDREQ	120	Requesting exchange 1; Asset 1; Reserved 30

Table B3: Packet layouts. Every packet is at most 1,024 bytes: the brief’s 64-byte header and at most 960 bytes of payload.

Unit	Bytes	Layout
Network header (brief)	64	Every packet; defined by the network
Data sub-header	16	Session 8, data sequence 4, record count 2, flags 2
Data packet	64 + 16 + records	Whole records, at most 944 bytes of records
SYN, SYN-ACK, ACK, DACK payload	96	Session 8, route 4, data sequence 4, attempt 1, kind 1, reserved 6, time 8, signature 64
Hop receipt payload	16	Packet ID 12, launch number 1, reserved 3
Direct sub-header	16	Principal 8, packet sequence 4, instruction count 1, flags 1, reserved 2
Direct packet	64 + 16 + instructions	Whole signed instructions, at most 944 bytes
FWD wrapper (R8)	24 + inner	Forwarder 1, target host 1, attempt 1, flags 1, forward time 8, origin principal 8, inner length 2, reserved 2

Table B4: Capacity. A data or direct packet carries a 16-byte sub-header and up to 944 bytes of whole records. “Packet bytes” is the full packet at that count, header included; it never exceeds 1,024. The last column is the number of client instructions that still fit when R8 wraps the batch in a 24-byte FWD header.

Record	Bytes	Per packet	Packet bytes	Per FWD packet
TRANSFER	176	5	960	—
RECEIVED	120	7	920	—
ORDER	128	7	976	7
CANCEL	128	7	976	7
WITHDRAW	128	7	976	7
CLOSE	128	7	976	7
CINSTR	256	3	848	3
PRICE	200	4	880	—
MDS	152	6	992	—
MDREQ	120	7	920	—
STATUS (52 entries)	944	1	1024	—

Packing and splitting rules. (1) A record is never split across packets. The sender takes records from the outbox for one peer in order and adds whole records while the payload stays within 960 bytes (16-byte sub-header plus at most 944 bytes of records); the next record starts a new packet. Every fixed record is at most 256 bytes, so up to 5 transfers, 7 receipts or 3 contract instructions fit in one packet (Table B4). The simulator’s packer asserts that every record is between 1 and 944 bytes. (2) STATUS is the only variable-length record. It is closed at 52 entries (944 bytes). If more remote clients need an entry at the same moment, the host writes further STATUS records, each a complete signed record of its own. (3) A client may batch whole instructions in one direct packet, up to 944 bytes. A larger batch is split into several direct packets, and each one counts against the client’s 12 per day. A batch that may need forwarding (R8) is limited to 920 bytes so that the 24-byte FWD header still fits; a forwarding exchange forwards each received packet as one FWD packet and never merges or splits them. The simulator asserts this limit on every direct send. (4) RECEIVED records are not sent alone if other records for the same peer are waiting: they share packets with whatever is in the outbox. A packet that holds only receipts uses the reserve quota (Section E6).

Examples. The S1a funding packet holds one TRANSFER: $64 + 16 + 176 = 256$ bytes. A backlog packet with five transfers is $64 + 16 + 880 = 960$ bytes. A STATUS record with one entry is 128 bytes and one with 52 entries is 944 bytes. The S1i batch of two orders is a direct packet of $64 + 16 + 256 = 336$ bytes, and a forwarded order is $64 + 16 + 24 + 128 = 232$ bytes.

S1+. Efficiency per customer

Per customer. The brief’s measures count every leg, so a \$100,000 purchase settles \$300,000 and costs 15.7 packets per transaction. Table B5 gives the same runs from the client’s side: one purchase with home delivery costs 47 packets and commits exactly its own price, and the two futures sides together post 80% of the notional (40% each). Funding a remote market moves the client’s own money to its own account there; the purchase is complete only when the shares are spendable at home, as the brief defines completion, and nothing remains encumbered afterwards. The funding leg is counted as a transaction because the brief counts completed transfers, but it is not a completed purchase. The component capital efficiency of 0.33 in Table A5 therefore reflects how the workflow is divided into legs; it does not mean that a client needs only a third of the price. Every purchase is fully prefunded.

Table B5: Efficiency per customer action (no loss). Value is the purchase price, or the notional for futures (both sides). Committed / value is the peak encumbered cash over that value; hours committed are dollar-hours over the peak.

Customer action	Value	Packets	Committed / value	Hours committed	Done (h)
Alice buys 1,000 Ares (Earth at Mars)	\$100,000	47	1.00	1.7	3.1
Alice buys 1,000 Belt (Earth at Ceres)	\$100,000	47	1.00	1.7	3.2
Eve buys 400 Terra (Uranus at Earth)	\$40,000	52	1.00	7.9	15.8
Futures, Q = 2,000, both sides	\$200,000	141	0.80	327.4	328.2
Repeat trading, \$10,000 funded	\$10,000	59	1.00	10.6	51.1

T. Complete traces

These tables give every row of every S1 run and of the three S3 relocations discussed in the text, in the brief’s seven-column format. Rows come from the simulator’s event stream. The financial state is the ledger snapshot immediately after the event, for the parties named in the caption: “free at host” is available at the market, “held” is an order or instruction hold, and “margin” is contract escrow. A local instruction is shown when it arrives at the exchange, 1 s after the client sends it (the send time is in the packet column), so that every row’s state is the state after that row. Packet columns list each backbone DATA launch of the record by hop; the handshake, acknowledgements and hop receipts of every session are in `traces/<run>_launches.csv`, whose counts are Table A4. The runs have no loss, so every first launch arrives.

Table B6: Complete trace of S1a: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Earth exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ea→Ma, batched with queued records	1.6966	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ea, batched with queued records	2.2622	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2622	Earth exchange	MDS from Mars	Posts Ares snapshot to its board (0.57 h old); answers waiting QUOTEs	Local access (1 s)	2.2625	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.57 h old	—	—	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Earth, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Earth	Session from h 0.0006; DATA Ma→A h 2.5027, A→Ea h 2.7549	3.0674	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5190	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5357	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.0674	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.6331	Alice \$50k at Earth, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
3.6331	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table B7: Complete trace of S1b: Alice (Earth) buys 1,000 Belt from Cara (Ceres) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Cara	Own inventory at the market	Sell 1,000 Belt limit \$100 GTC	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt home
0.0003	Ceres exchange	Its own available balances	Admits Cara EQB-sell; holds 1,000 Belt	—	—	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Alice	Listing table: Belt trades at Ceres	QUOTE for Belt	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Earth exchange	No fresh Belt snapshot on its board	Quote request (MDREQ) for Belt to Ceres	Backbone Ea→Ce, batched with queued records	1.6697	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Alice	Own Earth balance	Funds own Ceres account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held

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h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Ceres	New session (SYN h 0.0006); DATA Ea→A h 1.1138, A→Ce h 1.4269	1.6697	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres, 1,000 Belt held
1.6697	Ceres exchange	Its own live book	Replies with Belt snapshot (MDS, version 1)	Backbone Ce→Ea, batched with queued records	2.2263	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres, 1,000 Belt held
1.6697	Ceres exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2263	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2263	Earth exchange	MDS from Ceres	Posts Belt snapshot to its board (0.56 h old); answers waiting QUOTES	Local access (1 s)	2.2266	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2263	Earth exchange	Ceres imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2266	Alice	Home board (local access)	Reads Belt quote: bid none, ask \$100 × 1,000, 0.56 h old	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2266	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea→Ce, launches 2.2266, 2.2432, 2.2599; p(loss) 0.197 each, all lost 7.7e-03	2.6085 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.6085	Ceres exchange	Its own available balances	Admits Alice EQB-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held; Cara \$100k at Ceres, 1,000 Belt held
2.6085	Ceres exchange	Both holds in its own ledger	Fill 1,000 Belt @ \$100: Alice buys from Cara	—	local, atomic	Alice \$50k at Earth, 1,000 Belt at market; Cara \$200k at Ceres
2.6085	Ceres exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Belt; TRANSFER CE-X1 (automatic home delivery) to Earth	Session from h 0.0006; DATA Ce→A h 2.6087, A→Ea h 2.8520	3.1644	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres
2.6251	Ceres exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.6418	Ceres exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres
3.1644	Earth exchange	Authentic new transfer CE-X1	Imports once; credits Alice 1,000 Belt (spendable now)	RECEIVED to Ceres (backbone)	3.7210	Alice \$50k at Earth, 1,000 Belt home; Cara \$200k at Ceres
3.7210	Ceres exchange	Earth imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, 1,000 Belt home; Cara \$200k at Ceres

Table B8: Complete trace of S1c: Eve (Uranus) buys 400 Terra from Fin (Earth) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Fin	Own inventory at the market	Sell 400 Terra limit \$100 GTC	Local access at Earth, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 1,000 Terra home
0.0003	Earth exchange	Its own available balances	Admits Fin EQT-sell; holds 400 Terra	—	—	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Eve	Listing table: Terra trades at Earth	QUOTE for Terra	Local access at Uranus, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Uranus exchange	No fresh Terra snapshot on its board	Quote request (MDREQ) for Terra to Earth	Backbone Ur→Ea, batched with queued records	7.9134	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Eve	Own Uranus balance	Funds own Earth account: \$40k	Local access at Uranus, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Uranus exchange	Instruction or fill; its own ledger	Debits Eve \$40k; TRANSFER UR-X1 (funding) to Earth	New session (SYN h 0.0006); DATA Ur→A h 5.2765, A→Ea h 7.6012	7.9134	Eve \$10k at Uranus, \$40k in transit; Fin \$50k at Earth, 600 Terra home, 400 Terra held
7.9134	Earth exchange	Its own live book	Replies with Terra snapshot (MDS, version 1)	Backbone Ea→Ur, batched with queued records	10.5510	Eve \$10k at Uranus, \$40k in transit; Fin \$50k at Earth, 600 Terra home, 400 Terra held

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h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
7.9134	Earth exchange	Authentic new transfer UR-X1	Imports once; credits Eve \$40k (spendable now)	RECEIVED to Uranus (backbone)	10.5510	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5510	Uranus exchange	MDS from Earth	Posts Terra snapshot to its board (2.64 h old); answers waiting QUOTES	Local access (1 s)	10.5513	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5510	Uranus exchange	Earth imported UR-X1	Marks transfer delivered; no balance change	—	—	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5513	Eve	Home board (local access)	Reads Terra quote: bid none, ask \$100 × 400, 2.64 h old	—	—	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5513	Eve	Published geometry; own funding confirmed	Sends ORDER ×8 copies	Direct Ur→Ea, launches 10.5513, 10.5680, 10.5846, 10.6013, 10.6180, 10.6346, 10.6513, 10.6680; p(loss) 0.781 each, all lost 1.4e-01	13.1878 (first)	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
13.1878	Earth exchange	Its own available balances	Admits Eve EQT-buy; holds \$40k	—	—	Eve \$10k at Uranus, \$40k held; Fin \$50k at Earth, 600 Terra home, 400 Terra held
13.1878	Earth exchange	Both holds in its own ledger	Fill 400 Terra @ \$100: Eve buys from Fin	—	local, atomic	Eve \$10k at Uranus, 400 Terra at market; Fin \$90k at Earth, 600 Terra home
13.1878	Earth exchange	Instruction or fill; its own ledger	Debits Eve 400 Terra; TRANSFER EA-X1 (automatic home delivery) to Uranus	Session from h 0.0006; DATA Ea→A h 13.1881, A→Ur h 13.5004	15.8244	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2045	Earth exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
13.2211	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2378	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2545	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2711	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2878	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.3045	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
15.8244	Uranus exchange	Authentic new transfer EA-X1	Imports once; credits Eve 400 Terra (spendable now)	RECEIVED to Earth (backbone)	18.4611	Eve \$10k at Uranus, 400 Terra home; Fin \$90k at Earth, 600 Terra home
18.4611	Earth exchange	Uranus imported EA-X1	Marks transfer delivered; no balance change	—	—	Eve \$10k at Uranus, 400 Terra home; Fin \$90k at Earth, 600 Terra home

Table B9: Complete trace of S1d: capped futures, rising path, $Q = 2,000$ (Alice long, Cara short). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres

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h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held; Cara \$20k at Ceres, \$80k held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 110	Local access at Mars, sent h 62.5021	62.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 110	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 120	Local access at Mars, sent h 122.5021	122.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 120	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 130	Local access at Mars, sent h 182.5021	182.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 130	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 135	Local access at Mars, sent h 242.5021	242.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 135	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 125	Local access at Mars, sent h 302.5021	302.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 125	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$50k; gross long \$130k, short \$30k	—	settles; claims backed	Alice \$70k at Earth, \$130k free at host; Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$130k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$30k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$30k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$130k in transit; Cara \$50k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$130k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$200k at Earth; Cara \$50k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$200k at Earth; Cara \$50k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$200k at Earth; Cara \$50k at Ceres

Table B10: Complete trace of S1e: capped futures, falling path, $Q = 2,000$. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINST ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 90	Local access at Mars, sent h 62.5021	62.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 90	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 80	Local access at Mars, sent h 122.5021	122.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 80	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 70	Local access at Mars, sent h 182.5021	182.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 70	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 65	Local access at Mars, sent h 242.5021	242.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 65	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 75	Local access at Mars, sent h 302.5021	302.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 75	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 75: long net \$-50k; gross long \$30k, short \$130k	—	settles; claims backed	Alice \$70k at Earth, \$30k free at host; Cara \$20k at Ceres, \$130k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$30k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$70k at Earth, \$30k in transit; Cara \$20k at Ceres, \$130k free at host

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$130k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$70k at Earth, \$30k in transit; Cara \$20k at Ceres, \$130k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$130k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$30k in transit; Cara \$150k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$30k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$100k at Earth; Cara \$150k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$100k at Earth; Cara \$150k at Ceres
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$100k at Earth; Cara \$150k at Ceres

Table B11: Complete trace of S1f: capped futures, boundary case $Q = 2,500$. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$100k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$100k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$50k at Earth, \$100k in transit; Cara \$100k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$100k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$50k at Earth, \$100k in transit; Cara \$100k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR $\times 2$ copies	Direct Ce $\rightarrow$ Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$100k	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$100k	—	—	Alice \$50k at Earth, \$100k held; Cara \$100k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$200k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma $\rightarrow$ Ea	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 110	Local access at Mars, sent h 62.5021	62.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 110	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 120	Local access at Mars, sent h 122.5021	122.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 120	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 130	Local access at Mars, sent h 182.5021	182.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 130	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 135	Local access at Mars, sent h 242.5021	242.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 135	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 125	Local access at Mars, sent h 302.5021	302.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 125	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$62.5k; gross long \$162.5k, short \$37.5k	—	settles; claims backed	Alice \$50k at Earth, \$162.5k free at host; Cara \$37.5k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$162.5k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$37.5k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$37.5k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$162.5k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$212.5k at Earth; Cara \$37.5k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$212.5k at Earth; Cara \$37.5k at Ceres
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$212.5k at Earth; Cara \$37.5k at Ceres

Table B12: Complete trace of S1g: capped futures, $Q = 2,501$ (Cara cannot fund; Alice is refunded). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100.04k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100.04k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$100.04k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Ceres exchange	insufficient available ND: have 100000, need 100040	Rejects Cara FUND F1-Cara-fund	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100.04k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$100.04k	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma $\rightarrow$ Ea	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5190	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5357	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
3.0674	Earth exchange	STATUS from Mars: F1 HELD for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
168.0000	Mars exchange	opening deadline passed	Releases Alice margin \$100.04k	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
168.0000	Mars exchange	both instructions not held by deadline	Rejects F1	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
168.0000	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma $\rightarrow$ Ea	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
168.5604	Earth exchange	STATUS from Mars: F1 REJECTED (deadline) for Alice	Tells Alice (no balance change)	Local access (1 s)	168.5607	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.0000	Alice	no OPEN report by the opening deadline	withdraw after deadline	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.0000	Alice	Published geometry; own funding confirmed	Sends WITHDRAW $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 169.0000, 169.0167, 169.0333; p(loss) 0.126 each, all lost 2.0e-03	169.2336 (first)	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.2336	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$100.04k; TRANSFER MA-X1 (withdrawal) to Earth	Session from h 0.0006; DATA Ma $\rightarrow$ A h 169.2338, A $\rightarrow$ Ea h 169.4928	169.7939	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.2502	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.2669	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.7939	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$100.04k (spendable now)	RECEIVED to Mars (backbone)	170.3549	Alice \$150k at Earth; Cara \$100k at Ceres
170.3549	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$150k at Earth; Cara \$100k at Ceres

Table B13: Complete trace of S1h: capped futures, $Q = 4,000$ (both fundings refused at home). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$160k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	insufficient available ND: have 150000, need 160000	Rejects Alice FUND F1-Alice-fund	—	—	Alice \$150k at Earth; Cara \$100k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$160k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Ceres exchange	insufficient available ND: have 100000, need 160000	Rejects Cara FUND F1-Cara-fund	—	—	Alice \$150k at Earth; Cara \$100k at Ceres

Table B14: Complete trace of S1i: repeat trading on a \$10,000 Mars account. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Earth exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ea→Ma, batched with queued records	1.6966	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Alice	Own Earth balance	Funds own Mars account: \$10k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$10k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 3,000 Ares home
0.0003	Bob	Own inventory at the market	Sell 10 Ares limit \$97 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob RT-s1; holds 10 Ares	—	—	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,990 Ares home, 10 Ares held
0.0003	Bob	Own inventory at the market	Sell 5 Ares limit \$100 GTC	Local access at Mars, sent h 0.0001	0.0003	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,990 Ares home, 10 Ares held
0.0003	Mars exchange	Its own available balances	Admits Bob RT-s2; holds 5 Ares	—	—	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.6966	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 2)	Backbone Ma→Ea, batched with queued records	2.2622	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$10k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2622	Earth exchange	MDS from Mars	Posts Ares snapshot to its board (0.57 h old); answers waiting QUOTEs	Local access (1 s)	2.2625	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2625	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$97 × 10, 0.57 h old	—	—	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER o1+o2 ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.5024	Mars exchange	Its own available balances	Admits Alice RT-o1; holds \$4k	—	—	Alice \$140k at Earth, \$6k free at host, \$4k held; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 10 Ares @ \$97: Alice buys from Bob; improvement \$30	—	local, atomic	Alice \$140k at Earth, \$6.03k free at host, \$3k held, 10 Ares at market; Bob \$50.97k at Mars, 2,985 Ares home, 5 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 5 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5024	Mars exchange	Its own available balances	Admits Alice RT-o2; holds \$4k	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5190	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5190	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5357	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5357	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.2625	Alice	Published geometry; own funding confirmed	Sends CANCEL o2 ×3 copies	Direct Ea→Ma, launches 8.2625, 8.2791, 8.2958; p(loss) 0.129 each, all lost 2.1e-03	8.5021 (first)	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5021	Mars exchange	Order state	Cancelled RT-o2; releases \$4k	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5188	Mars exchange	RT-o2 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5355	Mars exchange	RT-o2 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
14.2625	Alice	Published geometry; own funding confirmed	Sends ORDER o3 ×3 copies	Direct Ea→Ma, launches 14.2625, 14.2791, 14.2958; p(loss) 0.129 each, all lost 2.1e-03	14.5019 (first)	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
14.5019	Mars exchange	Its own available balances	Admits Alice RT-o3; holds 10 Ares	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
14.5186	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
14.5353	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
15.2627	Bob	Own inventory at the market	Buy 10 Ares limit \$101 IOC	Local access at Mars, sent h 15.2625	15.2627	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
15.2627	Mars exchange	Its own available balances	Admits Bob RT-b1; holds \$1.01k	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$50.46k at Mars, \$1.01k held, 2,985 Ares home
15.2627	Mars exchange	Both holds in its own ledger	Fill 10 Ares @ \$101: Bob buys from Alice	—	local, atomic	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home
26.2625	Alice	Published geometry; own funding confirmed	Sends CANCEL o1 ×3 copies	Direct Ea→Ma, launches 26.2625, 26.2791, 26.2958; p(loss) 0.128 each, all lost 2.1e-03	26.5015 (first)	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home
26.3627	Bob	Own inventory at the market	Sell 5 Ares limit \$99 IOC	Local access at Mars, sent h 26.3625	26.3627	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home
26.3627	Mars exchange	Its own available balances	Admits Bob RT-s3; holds 5 Ares	—	—	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,990 Ares home, 5 Ares held
26.3627	Mars exchange	Both holds in its own ledger	Fill 5 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$140k at Earth, \$7.04k free at host, \$2k held, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
26.5015	Mars exchange	Order state	Cancelled RT-o1; releases \$2k	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
26.5181	Mars exchange	RT-o1 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
26.5348	Mars exchange	RT-o1 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.2625	Alice	Published geometry; own funding confirmed	Sends WITHDRAW cash+ARES $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 50.2625, 50.2791, 50.2958; p(loss) 0.128 each, all lost 2.1e-03	50.5006 (first)	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.5006	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$9.04k; TRANSFER MA-X1 (withdrawal) to Earth	Session from h 0.0006; DATA Ma $\rightarrow$ A h 50.5009, A $\rightarrow$ Ea h 50.7551	51.0642	Alice \$140k at Earth, \$9.04k in transit, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.5006	Mars exchange	Instruction or fill; its own ledger	Debits Alice 10 Ares; TRANSFER MA-X2 (withdrawal) to Earth	Session from h 0.0006; DATA Ma $\rightarrow$ A h 50.5009, A $\rightarrow$ Ea h 50.7551	51.0642	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5172	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5172	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5339	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5339	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
51.0642	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$9.04k (spendable now)	RECEIVED to Mars (backbone)	51.6284	Alice \$149.04k at Earth, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
51.0642	Earth exchange	Authentic new transfer MA-X2	Imports once; credits Alice 10 Ares (spendable now)	RECEIVED to Mars (backbone)	51.6284	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home
51.6284	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home
51.6284	Mars exchange	Earth imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home

E2+. Every direct message, every backbone launch and every run under loss

Table B15: Every direct message in every S1 run (exact). “First h” is the arrival of the first copy, the time used in the traces, and “P by then” the probability that the message has arrived by then. P(arrives) is the probability that at least one copy arrives, reached at the last copy’s arrival. S1h sends no direct message: both fundings are refused at home.

Run	Sender	Path	Message	Copies	p each	First h	P by then	Last h	P(arrives)
S1a	Alice	Ea→Ma	ORDER	3	0.1289	2.502	0.8711	2.536	0.997859
S1b	Alice	Ea→Ce	ORDER	3	0.1974	2.608	0.8026	2.642	0.992306
S1c	Eve	Ur→Ea	ORDER	8	0.7815	13.188	0.2185	13.304	0.860817
S1d	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1d	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1e	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1e	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1f	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1f	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1g	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1g	Alice	Ea→Ma	WITHDRAW	3	0.1257	169.234	0.8743	169.267	0.998015
S1i	Alice	Ea→Ma	ORDER o1 + o2	3	0.1289	2.502	0.8711	2.536	0.997859
S1i	Alice	Ea→Ma	CANCEL o2	3	0.1288	8.502	0.8712	8.535	0.997865
S1i	Alice	Ea→Ma	ORDER o3	3	0.1287	14.502	0.8713	14.535	0.997870
S1i	Alice	Ea→Ma	CANCEL o1	3	0.1284	26.501	0.8716	26.535	0.997881
S1i	Alice	Ea→Ma	WITHDRAW cash + ARES	3	0.1280	50.501	0.8720	50.534	0.997904

Table B16: Every backbone launch of every S1 run except hop receipts, with its per-launch probabilities. “p unconf.” is the chance that the launch is not confirmed: the data or its hop receipt is lost, $1 - (1 - p_{\text{data}})(1 - p_{\text{receipt}})$. P(aband.) is the chance that the hop is abandoned after four unconfirmed launches, $u_1 u_2 u_3 u_4$ with each retry at the hop timer and its own geometry; Bound holds only if no retry is delayed by more than 24 hours (Section E2). TR = TRANSFER, RCV = RECEIVED.

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1a	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1a	0.566	Ma	→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	0.818	A	→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.131	Ea	→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.132	Ea	→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.445	A	→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.445	A	→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.697	Ma	→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.697	Ma	→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.949	A	→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	1.950	A	→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	2.263	Ea	→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	2.503	Ma	→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	2.576	A	→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	2.755	A	→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.068	Ea	→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.068	Ea	→A	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.381	A	→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.381	A	→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.634	Ma	→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.886	A	→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	0.001	Ea	→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	0.314	A	→Ce	SYN	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	0.557	Ce	→A	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	0.800	A	→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.113	Ea	→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.114	Ea	→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.427	A	→Ce	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.427	A	→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.670	Ce	→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.670	Ce	→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.913	A	→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	1.914	A	→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	2.227	Ea	→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	2.540	A	→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	2.609	Ce	→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	2.852	A	→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.165	Ea	→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.165	Ea	→A	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.478	A	→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.478	A	→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.722	Ce	→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.965	A	→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	0.001	Ur	→A	SYN	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	2.325	A	→Ea	SYN	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	2.638	Ea	→A	SYNACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	2.951	A	→Ur	SYNACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	5.276	Ur	→A	ACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	5.277	Ur	→A	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	7.601	A	→Ea	ACK	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.601	A	→Ea	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.914	Ea	→A	DACK	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.914	Ea	→A	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1c	8.227	A→Ur	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	8.227	A→Ur	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	10.552	Ur→A	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	12.876	A→Ea	DACK	2.25	0.0440	0.0440	0.0860	5.5e-05	6.5e-05
S1c	13.188	Ea→A	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0860	5.5e-05	6.5e-05
S1c	13.500	A→Ur	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	15.825	Ur→A	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	15.825	Ur→A	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	18.149	A→Ea	DACK	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.150	A→Ea	DATA (RCV+TR)	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.462	Ea→A	DACK	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.774	A→Ur	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1d	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	0.001	Ce→A	SYN	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.244	A→Ma	SYN	1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1d	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.496	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.748	A→Ce	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	0.992	Ce→A	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.992	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	1.235	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.235	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.487	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.488	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.740	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.740	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	1.983	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.141	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.226	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.263	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	2.394	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.576	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.637	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.755	A→Ea	DATA (STATUS)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	2.755	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.880	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.999	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	3.068	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	3.242	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	3.381	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1d	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	326.768	A→Ea	SYN	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	326.768	A→Ce	SYN	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.013	Ce→A	SYNACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.060	Ea→A	SYNACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	327.257	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.351	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.523	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.523	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.617	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.617	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.788	A→Ce	ACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.788	A→Ce	DATA (RCV+TR)	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.882	A→Ea	ACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	327.882	A→Ea	DATA (RCV+TR)	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.033	Ce→A	DACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.033	Ce→A	DATA (RCV+TR)	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.174	Ea→A	DACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.174	Ea→A	DATA (RCV+TR)	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.277	A→Ma	DACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.277	A→Ma	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.466	A→Ma	DACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.466	A→Ma	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.543	Ma→A	DACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.731	Ma→A	DACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.808	A→Ce	DACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.997	A→Ea	DACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1e	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	0.001	Ce→A	SYN	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.244	A→Ma	SYN	1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1e	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	0.496	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	0.748	A→Ce	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	0.992	Ce→A	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.992	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	1.235	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.235	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.487	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.488	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.740	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.740	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1e	1.983	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.141	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.226	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	2.394	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.637	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.755	A→Ea	DATA (STATUS)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	2.755	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.880	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.999	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	3.068	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	3.242	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	3.381	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	326.768	A→Ea	SYN		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	326.768	A→Ce	SYN		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.013	Ce→A	SYNACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.060	Ea→A	SYNACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	327.257	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.351	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.523	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.523	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.617	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.617	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.788	A→Ce	ACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.788	A→Ce	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.882	A→Ea	ACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	327.882	A→Ea	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	328.033	Ce→A	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.033	Ce→A	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.174	Ea→A	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	328.174	Ea→A	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	328.277	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.277	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.466	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.466	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.543	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.731	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.808	A→Ce	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.997	A→Ea	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1f	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	0.001	Ce→A	SYN		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.244	A→Ma	SYN		1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1f	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.496	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.566	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.748	A→Ce	SYNACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.818	A→Ea	SYNACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1f	0.992	Ce→A	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.992	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	1.235	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.235	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.487	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.488	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.740	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.740	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	1.983	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.141	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.226	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.263	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	2.394	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.576	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.637	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.755	A→Ea	DATA (STATUS)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	2.755	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.880	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.999	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	3.068	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	3.242	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	3.381	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	326.768	A→Ea	SYN	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	326.768	A→Ce	SYN	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.013	Ce→A	SYNACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.060	Ea→A	SYNACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	327.257	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.351	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.523	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.523	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.617	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.617	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.788	A→Ce	ACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.788	A→Ce	DATA (RCV+TR)	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.882	A→Ea	ACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	327.882	A→Ea	DATA (RCV+TR)	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	328.033	Ce→A	DACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.033	Ce→A	DATA (RCV+TR)	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.174	Ea→A	DACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	328.174	Ea→A	DATA (RCV+TR)	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1f	328.277	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.277	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.466	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.466	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.543	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.731	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.808	A→Ce	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.997	A→Ea	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1g	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	0.566	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	0.818	A→Ea	SYNACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.131	Ea→A	ACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.132	Ea→A	DATA (RCV+TR)		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.445	A→Ma	ACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.445	A→Ma	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.697	Ma→A	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.697	Ma→A	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.949	A→Ea	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	1.950	A→Ea	DATA (RCV+TR)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	2.503	Ma→A	DATA (RCV+STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	2.755	A→Ea	DATA (RCV+STATUS)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	3.068	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	3.381	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	168.000	Ma→A	DATA (STATUS)		1.86	0.0366	0.0366	0.0718	2.7e-05	3.3e-05
S1g	168.259	A→Ea	DATA (STATUS)		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	168.561	Ea→A	DACK		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	168.863	A→Ma	DACK		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	169.234	Ma→A	DATA (TR)		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	169.493	A→Ea	DATA (TR)		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	169.794	Ea→A	DACK		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	169.795	Ea→A	DATA (RCV+STATUS)		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	170.096	A→Ma	DACK		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.096	A→Ma	DATA (RCV+STATUS)		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.355	Ma→A	DACK		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.614	A→Ea	DACK		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1i	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	0.566	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	0.818	A→Ea	SYNACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.131	Ea→A	ACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.132	Ea→A	DATA (RCV+TR)		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.445	A→Ma	ACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.445	A→Ma	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.697	Ma→A	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.697	Ma→A	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.949	A→Ea	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	1.950	A→Ea	DATA (RCV+TR)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1i	50.501	Ma→A	DATA (2×RCV+2×TR)	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	50.755	A→Ea	DATA (2×RCV+2×TR)	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.065	Ea→A	DACK	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.065	Ea→A	DATA (2×RCV+2×TR)	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.374	A→Ma	DACK	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.375	A→Ma	DATA (2×RCV+2×TR)	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.629	Ma→A	DACK	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.883	A→Ea	DACK	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05

Table B17: Per-run summary of Tables B15 and B16. The first probability is the chance that every backbone hop is confirmed at its first launch (direct copies not included); the second is the chance that at least one hop is abandoned and endpoint recovery takes over (exact, then the 24-hour-delay bound).

Run	Direct msgs	Backbone launches	P(no backbone retry)	P(some hop abandoned)	Bound
S1a	1	22	0.1668	8.7e-04	1.0e-03
S1b	1	22	0.1716	8.4e-04	1.0e-03
S1c	1	22	0.0002	4.8e-01	4.8e-01
S1d	2	68	0.0056	2.1e-03	2.5e-03
S1e	2	68	0.0056	2.1e-03	2.5e-03
S1f	2	68	0.0056	2.1e-03	2.5e-03
S1g	2	30	0.0877	1.2e-03	1.4e-03
S1h	0	0	1.0000	0.0e+00	0.0e+00
S1i	5	22	0.1671	8.7e-04	1.0e-03

Table B18: Monte Carlo under random loss for every S1 run with network traffic (2,000 runs each). The last column is the 95% upper confidence bound on the probability of not reaching the end point within the run (1,500 hours) when no run failed (rule of three, $3/n$). S1h has no network traffic.

Run	End point	Done	Median h	p90 h	p99 h	Max h	95\% bound
S1a	completion	2000/2000	3.07	4.71	7.70	28.21	0.15%
S1b	completion	2000/2000	3.16	4.81	7.76	14.04	0.15%
S1c	completion	2000/2000	28.76	50.74	71.48	104.58	0.15%
S1d	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1e	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1f	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1g	margin home	2000/2000	169.79	170.16	171.77	177.40	0.15%
S1i	completion	2000/2000	51.06	52.71	55.70	77.70	0.15%

E3+. State histories and settlement moments

Table A13 is the opening balance sheet. For each scenario the next tables list every change in the positions of its two parties, one row per account and asset that changed, so that the whole ledger history can be read off without the trace files. “Home” is available at the home exchange, “At host” is available in the remote trading account, “Locked” is order and instruction holds plus contract margin, and “Transit” is value debited at one exchange and not yet credited at another. All other accounts are unchanged in every run.

Table B19: State history of S1a: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQ-sell	Bob	Ares	2,000	0	1,000	0
0.0003	Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
1.6966	Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
2.5024	Order admitted EQ-buy	Alice	ND	\$50,000	0	\$100,000	0

(continued)

h	Event	Account	Asset	Home	At host	Locked	Transit
2.5024	Fill 1,000 Ares @ \$100	Alice	ND	\$50,000	0	0	0
2.5024	Fill 1,000 Ares @ \$100	Alice	Ares	0	1,000	0	0
2.5024	Fill 1,000 Ares @ \$100	Bob	ND	\$150,000	0	0	0
2.5024	Fill 1,000 Ares @ \$100	Bob	Ares	2,000	0	0	0
2.5024	Export MA-X1 (auto fill)	Alice	Ares	0	0	0	1,000
3.0674	Import MA-X1 (auto fill)	Alice	Ares	1,000	0	0	0

Table B20: State history of S1b: Alice (Earth) buys 1,000 Belt from Cara (Ceres) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQB-sell	Cara	Belt	0	0	1,000	0
0.0003	Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
1.6697	Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
2.6085	Order admitted EQB-buy	Alice	ND	\$50,000	0	\$100,000	0
2.6085	Fill 1,000 Belt @ \$100	Alice	ND	\$50,000	0	0	0
2.6085	Fill 1,000 Belt @ \$100	Alice	Belt	0	1,000	0	0
2.6085	Fill 1,000 Belt @ \$100	Cara	ND	\$200,000	0	0	0
2.6085	Fill 1,000 Belt @ \$100	Cara	Belt	0	0	0	0
2.6085	Export CE-X1 (auto fill)	Alice	Belt	0	0	0	1,000
3.1644	Import CE-X1 (auto fill)	Alice	Belt	1,000	0	0	0

Table B21: State history of S1c: Eve (Uranus) buys 400 Terra from Fin (Earth) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQT-sell	Fin	Terra	600	0	400	0
0.0003	Export UR-X1 (fund)	Eve	ND	\$10,000	0	0	\$40,000
7.9134	Import UR-X1 (fund)	Eve	ND	\$10,000	\$40,000	0	0
13.1878	Order admitted EQT-buy	Eve	ND	\$10,000	0	\$40,000	0
13.1878	Fill 400 Terra @ \$100	Eve	ND	\$10,000	0	0	0
13.1878	Fill 400 Terra @ \$100	Eve	Terra	0	400	0	0
13.1878	Fill 400 Terra @ \$100	Fin	ND	\$90,000	0	0	0
13.1878	Fill 400 Terra @ \$100	Fin	Terra	600	0	0	0
13.1878	Export EA-X1 (auto fill)	Eve	Terra	0	0	0	400
15.8244	Import EA-X1 (auto fill)	Eve	Terra	400	0	0	0

Table B22: State history of S1d: capped futures, rising path, $Q = 2,000$ (Alice long, Cara short).

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Export EA-X1 (fund)	Alice	ND	\$70,000	0	0	\$80,000
0.0003	Export CE-X1 (fund)	Cara	ND	\$20,000	0	0	\$80,000
1.4868	Import CE-X1 (fund)	Cara	ND	\$20,000	\$80,000	0	0
1.6966	Import EA-X1 (fund)	Alice	ND	\$70,000	\$80,000	0	0
2.1410	Margin held F1	Cara	ND	\$20,000	0	\$80,000	0
2.5024	Margin held F1	Alice	ND	\$70,000	0	\$80,000	0
326.5024	Contract fixed F1	Alice	ND	\$70,000	\$130,000	0	0
326.5024	Contract fixed F1	Cara	ND	\$20,000	\$30,000	0	0
326.5024	Export MA-X1 (payout)	Alice	ND	\$70,000	0	0	\$130,000
326.5024	Export MA-X2 (payout)	Cara	ND	\$20,000	0	0	\$30,000
328.0321	Import MA-X2 (payout)	Cara	ND	\$50,000	0	0	0
328.1736	Import MA-X1 (payout)	Alice	ND	\$200,000	0	0	0

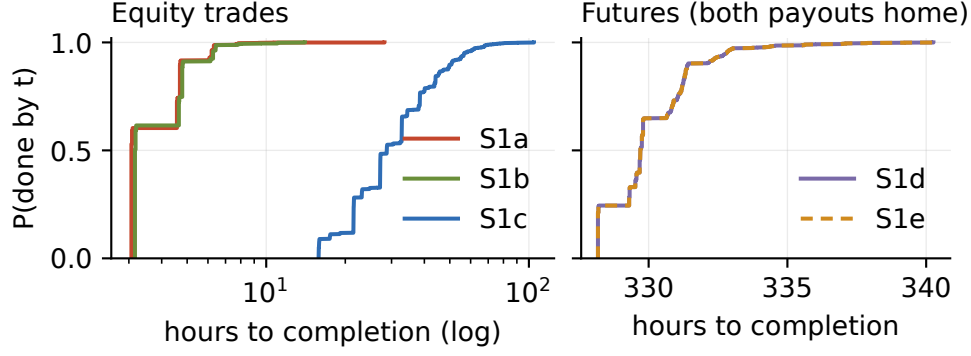


Figure B1: Completion-time distributions under random loss (empirical CDF of the Monte Carlo runs in Table A10). Left: the three equity trades, log scale. Right: the futures, both payouts home.

Table B23: State history of S1e: capped futures, falling path, $Q = 2,000$.

	h	Event	Account	Asset	Home	At host	Locked	Transit
	0.0003	Export EA-X1 (fund)	Alice	ND	\$70,000	0	0	\$80,000
	0.0003	Export CE-X1 (fund)	Cara	ND	\$20,000	0	0	\$80,000
	1.4868	Import CE-X1 (fund)	Cara	ND	\$20,000	\$80,000	0	0
	1.6966	Import EA-X1 (fund)	Alice	ND	\$70,000	\$80,000	0	0
	2.1410	Margin held F1	Cara	ND	\$20,000	0	\$80,000	0
	2.5024	Margin held F1	Alice	ND	\$70,000	0	\$80,000	0
	326.5024	Contract fixed F1	Alice	ND	\$70,000	\$30,000	0	0
	326.5024	Contract fixed F1	Cara	ND	\$20,000	\$130,000	0	0
	326.5024	Export MA-X1 (payout)	Alice	ND	\$70,000	0	0	\$30,000
	326.5024	Export MA-X2 (payout)	Cara	ND	\$20,000	0	0	\$130,000
	328.0321	Import MA-X2 (payout)	Cara	ND	\$150,000	0	0	0
	328.1736	Import MA-X1 (payout)	Alice	ND	\$100,000	0	0	0

Table B24: State history of S1f: capped futures, boundary case $Q = 2,500$.

	h	Event	Account	Asset	Home	At host	Locked	Transit
	0.0003	Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
	0.0003	Export CE-X1 (fund)	Cara	ND	0	0	0	\$100,000
	1.4868	Import CE-X1 (fund)	Cara	ND	0	\$100,000	0	0
	1.6966	Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
	2.1410	Margin held F1	Cara	ND	0	0	\$100,000	0
	2.5024	Margin held F1	Alice	ND	\$50,000	0	\$100,000	0
	326.5024	Contract fixed F1	Alice	ND	\$50,000	\$162,500	0	0
	326.5024	Contract fixed F1	Cara	ND	0	\$37,500	0	0
	326.5024	Export MA-X1 (payout)	Alice	ND	\$50,000	0	0	\$162,500
	326.5024	Export MA-X2 (payout)	Cara	ND	0	0	0	\$37,500
	328.0321	Import MA-X2 (payout)	Cara	ND	\$37,500	0	0	0
	328.1736	Import MA-X1 (payout)	Alice	ND	\$212,500	0	0	0

Table B25: State history of S1g: capped futures, $Q = 2,501$ (Cara cannot fund; Alice is refunded).

	h	Event	Account	Asset	Home	At host	Locked	Transit
	0.0003	Export EA-X1 (fund)	Alice	ND	\$49,960	0	0	\$100,040
	1.6966	Import EA-X1 (fund)	Alice	ND	\$49,960	\$100,040	0	0
	2.5024	Margin held F1	Alice	ND	\$49,960	0	\$100,040	0
	168.0000	Margin released F1	Alice	ND	\$49,960	\$100,040	0	0
	169.2336	Export MA-X1 (withdraw)	Alice	ND	\$49,960	0	0	\$100,040

(continued)

h	Event	Account	Asset	Home	At host	Locked	Transit
169.7939	Import MA-X1 (withdraw)	Alice	ND	\$150,000	0	0	0

Table B26: State history of S1i: repeat trading on a \$10,000 Mars account.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Export EA-X1 (fund)	Alice	ND	\$140,000	0	0	\$10,000
0.0003	Order admitted RT-s1	Bob	Ares	2,990	0	10	0
0.0003	Order admitted RT-s2	Bob	Ares	2,985	0	15	0
1.6966	Import EA-X1 (fund)	Alice	ND	\$140,000	\$10,000	0	0
2.5024	Order admitted RT-o1	Alice	ND	\$140,000	\$6,000	\$4,000	0
2.5024	Fill 10 Ares @ \$97	Alice	ND	\$140,000	\$6,030	\$3,000	0
2.5024	Fill 10 Ares @ \$97	Alice	Ares	0	10	0	0
2.5024	Fill 10 Ares @ \$97	Bob	ND	\$50,970	0	0	0
2.5024	Fill 10 Ares @ \$97	Bob	Ares	2,985	0	5	0
2.5024	Fill 5 Ares @ \$100	Alice	ND	\$140,000	\$6,030	\$2,500	0
2.5024	Fill 5 Ares @ \$100	Alice	Ares	0	15	0	0
2.5024	Fill 5 Ares @ \$100	Bob	ND	\$51,470	0	0	0
2.5024	Fill 5 Ares @ \$100	Bob	Ares	2,985	0	0	0
2.5024	Order admitted RT-o2	Alice	ND	\$140,000	\$2,030	\$6,500	0
8.5021	Cancelled RT-o2	Alice	ND	\$140,000	\$6,030	\$2,500	0
14.5019	Order admitted RT-o3	Alice	Ares	0	5	10	0
15.2627	Order admitted RT-b1	Bob	ND	\$50,460	0	\$1,010	0
15.2627	Fill 10 Ares @ \$101	Alice	ND	\$140,000	\$7,040	\$2,500	0
15.2627	Fill 10 Ares @ \$101	Alice	Ares	0	5	0	0
15.2627	Fill 10 Ares @ \$101	Bob	ND	\$50,460	0	0	0
15.2627	Fill 10 Ares @ \$101	Bob	Ares	2,995	0	0	0
26.3627	Order admitted RT-s3	Bob	Ares	2,990	0	5	0
26.3627	Fill 5 Ares @ \$100	Alice	ND	\$140,000	\$7,040	\$2,000	0
26.3627	Fill 5 Ares @ \$100	Alice	Ares	0	10	0	0
26.3627	Fill 5 Ares @ \$100	Bob	ND	\$50,960	0	0	0
26.3627	Fill 5 Ares @ \$100	Bob	Ares	2,990	0	0	0
26.5015	Cancelled RT-o1	Alice	ND	\$140,000	\$9,040	0	0
50.5006	Export MA-X1 (withdraw)	Alice	ND	\$140,000	0	0	\$9,040
50.5006	Export MA-X2 (withdraw)	Alice	Ares	0	0	0	10
51.0642	Import MA-X1 (withdraw)	Alice	ND	\$149,040	0	0	0
51.0642	Import MA-X2 (withdraw)	Alice	Ares	10	0	0	0

Settlement moments. Table B27 lists every obligation in every scenario. An obligation is *discharged* when the duty it records is met (a fill, a fixing, or a transfer credited at its destination); a claim is *backed* once the value has been debited into a transfer, so nobody can spend it twice; and it is *spendable* when the owner can use it, either at the host or at home.

Table B27: Settlement moments of every obligation in every S1 scenario (hours, no loss).

Run	Obligation	Created h	Discharged h	Backed	Spendable
S1a	Fund \$100,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1a	Fill 1,000 Ares @ \$100: Alice from Bob	2.502	2.502	local	2.502 at host
S1a	Auto Fill 1,000 Ares for Alice, Ma→Ea	2.502	3.067	2.502	3.067 at home
S1b	Fund \$100,000 for Alice, Ea→Ce	0.000	1.670	0.000	1.670 at host
S1b	Fill 1,000 Belt @ \$100: Alice from Cara	2.608	2.608	local	2.608 at host

(continued)

Run	Obligation	Created h	Discharged h	Backed	Spendable
S1b	Auto Fill 1,000 Belt for Alice, Ce→Ea	2.608	3.164	2.608	3.164 at home
S1c	Fund \$40,000 for Eve, Ur→Ea	0.000	7.913	0.000	7.913 at host
S1c	Fill 400 Terra @ \$100: Eve from Fin	13.188	13.188	local	13.188 at host
S1c	Auto Fill 400 Terra for Eve, Ea→Ur	13.188	15.824	13.188	15.824 at home
S1d	Fund \$80,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1d	Fund \$80,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1d	Contract F1 (escrow \$160,000)	2.502	at fixing	escrow from 2.502	—
S1d	Fixing of F1 at 125	326.502	326.502	326.502 (payouts)	—
S1d	Payout \$130,000 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1d	Payout \$30,000 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1e	Fund \$80,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1e	Fund \$80,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1e	Contract F1 (escrow \$160,000)	2.502	at fixing	escrow from 2.502	—
S1e	Fixing of F1 at 75	326.502	326.502	326.502 (payouts)	—
S1e	Payout \$30,000 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1e	Payout \$130,000 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1f	Fund \$100,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1f	Fund \$100,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1f	Contract F1 (escrow \$200,000)	2.502	at fixing	escrow from 2.502	—
S1f	Fixing of F1 at 125	326.502	326.502	326.502 (payouts)	—
S1f	Payout \$162,500 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1f	Payout \$37,500 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1g	Fund \$100,040 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1g	Withdraw \$100,040 for Alice, Ma→Ea	169.234	169.794	169.234	169.794 at home
S1i	Fund \$10,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1i	Fill 10 Ares @ \$97: Alice from Bob	2.502	2.502	local	2.502 at host
S1i	Fill 5 Ares @ \$100: Alice from Bob	2.502	2.502	local	2.502 at host
S1i	Fill 10 Ares @ \$101: Bob from Alice	15.263	15.263	local	15.263 at host
S1i	Fill 5 Ares @ \$100: Alice from Bob	26.363	26.363	local	26.363 at host
S1i	Withdraw \$9,040 for Alice, Ma→Ea	50.501	51.064	50.501	51.064 at home
S1i	Withdraw 10 Ares for Alice, Ma→Ea	50.501	51.064	50.501	51.064 at home

S2+. Incident search and complete trace

We chose the endpoint reset of the Mars exchange at hour 0.75, which is the worst incident on both price paths. It does more damage than the obvious choice, isolating the host at payout time (about 75 hours), even though no party can observe it. At hour 0.75, Mars has just answered Alice's and Cara's handshakes. The reset erases its session state, but its SYN-ACK replies are already in flight. Earth and Ceres therefore believe that the sessions are open and send their funding transfers into sessions that Mars no longer knows. As the

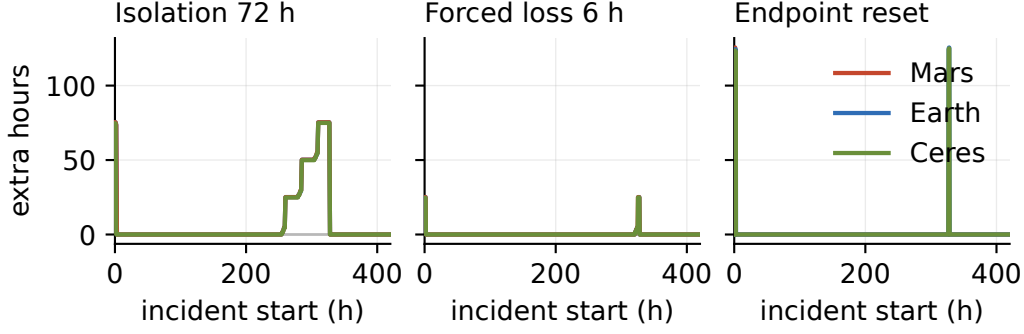


Figure B2: Damage (extra hours until both payouts are home, capped at 200) against incident start time on the rising path. Coloured lines are the three exchanges in the scenario; grey lines are the other nodes.

brief requires, Mars ignores them. The senders learn nothing until four endpoint attempts of $R_e \approx 25$ hours have expired. They then wait for the 24-hour application backoff and resubmit. The courier log of the rising path shows this sequence. Earth and Ceres open their sessions with Mars at hour 0.0003. Ceres declares its delivery unknown at hour 100.97 and Earth at hour 101.64. Each schedules an application resubmission 24 hours later, and they resubmit in new sessions at hours 124.97 and 125.64.

Complete trace of the incident. Table B28 lists every event of the selected incident up to the opening; the files `traces/S2_reset_Mars_{launches,transport,ledger}.csv` hold the whole run, including the 300-hour contract and the payouts, which then follow S1d shifted by the delay. The table shows the four steps of the 125-hour penalty: the handshakes complete before the reset, so Earth and Ceres send their transfers into sessions Mars has forgotten; Mars ignores every copy; each sender’s four endpoint attempts expire (delivery unknown); and after the 24-hour application backoff each resubmits on a new session, which succeeds within about two hours.

Table B28: Event-by-event trace of the S2 incident (rising path, no loss) from hour 0 to the opening at 128.126: every backbone launch except the 66 hop receipts, every direct copy, every courier event and every ledger event. Ea = Earth, Ma = Mars, Ce = Ceres, A = Relay A.

h	At	Event	Detail
0.000	Earth	session open	session s26 to Mars via Earth, Relay A, Mars
0.000	Ceres	session open	session s27 to Mars via Ceres, Relay A, Mars
0.000	Alice	local instruction	FUND \$80,000 to Mars
0.000	Earth	export	EA-X1: \$80,000 of Alice debited, in transit to Mars
0.000	Cara	local instruction	FUND \$80,000 to Mars
0.000	Ceres	export	CE-X1: \$80,000 of Cara debited, in transit to Mars
0.001	Ce→A	launch	SYN s27, arrives 0.243
0.001	Ea→A	launch	SYN s26, arrives 0.313
0.244	A→Ma	launch	SYN s27, arrives 0.495
0.314	A→Ma	launch	SYN s26, arrives 0.565
0.496	Ma→A	launch	SYNACK s27, arrives 0.748
0.566	Ma→A	launch	SYNACK s26, arrives 0.818
0.748	A→Ce	launch	SYNACK s27, arrives 0.991
0.750	Mars	endpoint reset	all session state lost; ledger and durable outbox kept
0.818	A→Ea	launch	SYNACK s26, arrives 1.131
0.992	Ce→A	launch	ACK s27, arrives 1.234
0.992	Ce→A	launch	DATA s27 (carries DATA), arrives 1.235
1.131	Ea→A	launch	ACK s26, arrives 1.444
1.132	Ea→A	launch	DATA s26 (carries DATA), arrives 1.444
1.235	A→Ma	launch	ACK s27, arrives 1.487
1.235	A→Ma	launch	DATA s27 (carries DATA), arrives 1.487
1.445	A→Ma	launch	ACK s26, arrives 1.696
1.445	A→Ma	launch	DATA s26 (carries DATA), arrives 1.697

(continued)

h	At	Event	Detail
1.487	Mars	ignored	ACK on s27 from Ceres: Mars lost this session in the reset
1.487	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
1.696	Mars	ignored	ACK on s26 from Earth: Mars lost this session in the reset
1.697	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
25.982	Ce→A	launch	DATA s27 (carries DATA), arrives 26.225
26.225	A→Ma	launch	DATA s27 (carries DATA), arrives 26.478
26.261	Ea→A	launch	DATA s26 (carries DATA), arrives 26.572
26.478	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
26.573	A→Ma	launch	DATA s26 (carries DATA), arrives 26.826
26.826	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
50.974	Ce→A	launch	DATA s27 (carries DATA), arrives 51.217
51.218	A→Ma	launch	DATA s27 (carries DATA), arrives 51.472
51.390	Ea→A	launch	DATA s26 (carries DATA), arrives 51.699
51.472	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
51.700	A→Ma	launch	DATA s26 (carries DATA), arrives 51.954
51.954	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
75.969	Ce→A	launch	DATA s27 (carries DATA), arrives 76.212
76.213	A→Ma	launch	DATA s27 (carries DATA), arrives 76.467
76.467	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
76.518	Ea→A	launch	DATA s26 (carries DATA), arrives 76.825
76.826	A→Ma	launch	DATA s26 (carries DATA), arrives 77.080
77.080	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
100.966	Ceres	delivery unknown	s27 to Mars: four endpoint attempts unacknowledged
100.966	Ceres	app resubmit scheduled	resubmission of the transfer scheduled for 124.966
101.643	Earth	delivery unknown	s26 to Mars: four endpoint attempts unacknowledged
101.643	Earth	app resubmit scheduled	resubmission of the transfer scheduled for 125.643
124.966	Ceres	app resubmit	transfer resubmitted to Mars on a new session
124.966	Ceres	session open	session s28 to Mars via Ceres, Relay A, Mars
124.966	Ce→A	launch	SYN s28, arrives 125.209
125.210	A→Ma	launch	SYN s28, arrives 125.467
125.467	Ma→A	launch	SYNACK s28, arrives 125.724
125.643	Earth	app resubmit	transfer resubmitted to Mars on a new session
125.643	Earth	session open	session s29 to Mars via Earth, Relay A, Mars
125.643	Ea→A	launch	SYN s29, arrives 125.948
125.724	A→Ce	launch	SYNACK s28, arrives 125.967
125.948	A→Ma	launch	SYN s29, arrives 126.205
125.968	Ce→A	launch	ACK s28, arrives 126.211
125.968	Ce→A	launch	DATA s28 (carries DATA), arrives 126.211
126.205	Ma→A	launch	SYNACK s29, arrives 126.462
126.212	A→Ma	launch	ACK s28, arrives 126.468
126.212	A→Ma	launch	DATA s28 (carries DATA), arrives 126.469
126.463	A→Ea	launch	SYNACK s29, arrives 126.767
126.469	Mars	import	CE-X1: \$80,000 credited to Cara

(continued)

h	At	Event	Detail
126.469	Mars	received queued	RECEIVED for CE-X1 queued to Ceres
126.469	Ma→A	launch	DACK s28, arrives 126.726
126.470	Ma→A	launch	DATA s28 (carries DATA), arrives 126.726
126.726	A→Ce	launch	DACK s28, arrives 126.970
126.727	A→Ce	launch	DATA s28 (carries DATA), arrives 126.970
126.767	Ea→A	launch	ACK s29, arrives 127.071
126.767	Ea→A	launch	DATA s29 (carries DATA), arrives 127.071
126.970	Ceres	receipt known	CE-X1 resolved
126.970	Ce→A	launch	DACK s28, arrives 127.214
126.971	Ce→Ma	launch	direct CINSTR copy 1 from Cara, arrives 127.127
126.987	Ce→Ma	launch	direct CINSTR copy 2 from Cara, arrives 127.143
127.072	A→Ma	launch	ACK s29, arrives 127.328
127.072	A→Ma	launch	DATA s29 (carries DATA), arrives 127.329
127.127	Mars	margin held	Cara: \$80,000 margin held
127.127	Mars	status queued	STATUS record to Ceres
127.127	Ma→A	launch	DATA s28 (carries DATA), arrives 127.384
127.143	Mars	duplicate cinstr	later copy from Cara ignored
127.214	A→Ma	launch	DACK s28, arrives 127.471
127.329	Mars	import	EA-X1: \$80,000 credited to Alice
127.329	Mars	received queued	RECEIVED for EA-X1 queued to Earth
127.329	Ma→A	launch	DACK s29, arrives 127.586
127.329	Ma→A	launch	DATA s29 (carries DATA), arrives 127.586
127.384	A→Ce	launch	DATA s28 (carries DATA), arrives 127.627
127.586	A→Ea	launch	DACK s29, arrives 127.890
127.587	A→Ea	launch	DATA s29 (carries DATA), arrives 127.891
127.627	Ceres	status received	Cara told HELD
127.628	Ce→A	launch	DACK s28, arrives 127.871
127.872	A→Ma	launch	DACK s28, arrives 128.128
127.891	Earth	receipt known	EA-X1 resolved
127.891	Ea→A	launch	DACK s29, arrives 128.195
127.891	Ea→Ma	launch	direct CINSTR copy 1 from Alice, arrives 128.126
127.908	Ea→Ma	launch	direct CINSTR copy 2 from Alice, arrives 128.143
127.925	Ea→Ma	launch	direct CINSTR copy 3 from Alice, arrives 128.159
128.126	Mars	margin held	Alice: \$80,000 margin held
128.126	Mars	contract open	F1 opens, escrow \$160,000
128.126	Mars	status queued	STATUS record to Earth
128.126	Mars	status queued	STATUS record to Ceres
128.126	Ma→A	launch	DATA s29 (carries DATA), arrives 128.383
128.127	Ma→A	launch	DATA s28 (carries DATA), arrives 128.383
128.143	Mars	duplicate cinstr	later copy from Alice ignored
128.159	Mars	duplicate cinstr	later copy from Alice ignored
128.196	A→Ma	launch	DACK s29, arrives 128.452
128.384	A→Ea	launch	DATA s29 (carries DATA), arrives 128.688
128.384	A→Ce	launch	DATA s28 (carries DATA), arrives 128.627
128.627	Ceres	status received	Cara told OPEN
128.628	Ce→A	launch	DACK s28, arrives 128.871
128.688	Earth	status received	Alice told OPEN
128.688	Ea→A	launch	DACK s29, arrives 128.992
128.871	A→Ma	launch	DACK s28, arrives 129.128
128.992	A→Ma	launch	DACK s29, arrives 129.249

S3+. Complete relocation traces

S3 relocations. The same \$100,000 Ares purchase with Alice’s account moved to the best, median and worst homes of Table A19.

Table B29: Complete trace of the S3 relocation, best home: Alice at Ceres buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Ceres exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ce→Ma, batched with queued records	1.4868	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Ceres balance	Funds own Mars account: \$100k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$50k at Ceres, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.4868	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ce, batched with queued records	1.9826	Alice \$50k at Ceres, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9826	Ceres exchange	MDS from Mars	Posts Ares snapshot to its board (0.50 h old); answers waiting QUOTEs	Local access (1 s)	1.9829	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.9829	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.50 h old	—	—	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9829	Alice	Published geometry; own funding confirmed	Sends ORDER ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1410	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Ceres, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1410	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Ceres, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
2.1410	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Ceres	Session from h 0.0006; DATA Ma→A h 2.1412, A→Ce h 2.3935	2.6363	Alice \$50k at Ceres, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.1576	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Ceres, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.6363	Ceres exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.1321	Alice \$50k at Ceres, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
3.1321	Mars exchange	Ceres imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Ceres, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table B30: Complete trace of the S3 relocation, median home: Alice at Mercury buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Mercury, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Mercury exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Me→Ma, batched with queued records	2.1195	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Mercury balance	Funds own Mars account: \$100k	Local access at Mercury, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Mercury exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER ME-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Me→A h 1.4136, A→Ma h 1.8678	2.1195	Alice \$50k at Mercury, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1195	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Me, batched with queued records	2.8262	Alice \$50k at Mercury, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1195	Mars exchange	Authentic new transfer ME-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Mercury (backbone)	2.8262	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8262	Mercury exchange	MDS from Mars	Posts Ares snapshot to its board (0.71 h old); answers waiting QUOTES	Local access (1 s)	2.8264	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8262	Mercury exchange	Mars imported ME-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8264	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.71 h old	—	—	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8264	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Me→Ma, launches 2.8264, 2.8431, 2.8598; p(loss) 0.147 each, all lost 3.2e-03	3.1035 (first)	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
3.1035	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Mercury, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
3.1035	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Mercury, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
3.1035	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Mercury	Session from h 0.0006; DATA Ma→A h 3.1037, A→Me h 3.3561	3.8096	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.1201	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.1368	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.8096	Mercury exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	4.5163	Alice \$50k at Mercury, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
4.5163	Mars exchange	Mercury imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Mercury, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table B31: Complete trace of the S3 relocation, worst home: Alice at Neptune buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Neptune, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Neptune exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ne→Ma, batched with queued records	12.3441	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Neptune balance	Funds own Mars account: \$100k	Local access at Neptune, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Neptune exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER NE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ne→A h 8.2296, A→Ma h 12.0919	12.3441	Alice \$50k at Neptune, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
12.3441	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ne, batched with queued records	16.4594	Alice \$50k at Neptune, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
12.3441	Mars exchange	Authentic new transfer NE-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Neptune (backbone)	16.4594	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4594	Neptune exchange	MDS from Mars	Posts Ares snapshot to its board (4.12 h old); answers waiting QUOTEs	Local access (1 s)	16.4597	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4594	Neptune exchange	Mars imported NE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4597	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 4.12 h old	—	—	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4597	Alice	Published geometry; own funding confirmed	Sends ORDER ×8 copies	Direct Ne→Ma, launches 16.4597, 16.4763, 16.4930, 16.5097, 16.5263, 16.5430, 16.5597, 16.5763; p(loss) 0.906 each, all lost 4.6e-01	20.5653 (first)	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
20.5653	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Neptune, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
20.5653	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Neptune, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
20.5653	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Neptune	Session from h 0.0006; DATA Ma→A h 20.5656, A→Ne h 20.8186	24.6808	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.5820	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.5987	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6153	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6320	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6487	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6653	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6820	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
24.6808	Neptune exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	28.7974	Alice \$50k at Neptune, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
28.7974	Mars exchange	Neptune imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Neptune, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

E5+. Every direct leg at the shifted epochs

Which path each instruction took. Table B32 lists every direct leg in the shifted-epoch runs of S1a and S1d. At +1 year and +100 years every instruction goes direct. R8 forwards instructions at +10 years and Mars conjunction, through the exchange named in the Sender column of the second leg.

Table B32: Every direct leg in the E5 runs of S1a and S1d (no loss). A forwarded instruction has two legs: the client to the forwarding exchange and that exchange to the host; it arrives with the product of the two P(arrives). Leg: fwd 1 is client to forwarder, fwd 2 is forwarder to host.

Epoch	Run	Sender	Path	Leg	Message	Copies	p	each	First	h	P(arrives)
+1 year	S1a	Alice	Ea→Ma	direct	ORDER	3	0.151	3.43	0.9966		
+1 year	S1d	Cara	Ce→Ma	direct	CINSTR	3	0.196	2.51	0.9925		
+1 year	S1d	Alice	Ea→Ma	direct	CINSTR	3	0.151	3.43	0.9966		
+10 years	S1a	Alice	Ea→Me	fwd 1	ORDER	2	0.067	2.92	0.9956		
+10 years	S1a	Mercury (R8)	Me→Ma	fwd 2	ORDER	3	0.137	3.17	0.9974		
+10 years	S1d	Cara	Ce→Ma	direct	CINSTR	3	0.102	1.44	0.9989		
+10 years	S1d	Alice	Ea→Me	fwd 1	CINSTR	2	0.067	2.92	0.9956		
+10 years	S1d	Mercury (R8)	Me→Ma	fwd 2	CINSTR	3	0.137	3.17	0.9974		
+100 years	S1a	Alice	Ea→Ma	direct	ORDER	3	0.162	2.88	0.9957		
+100 years	S1d	Alice	Ea→Ma	direct	CINSTR	3	0.162	2.88	0.9957		
+100 years	S1d	Cara	Ce→Ma	direct	CINSTR	4	0.296	3.94	0.9923		
Mars conjunction	S1a	Alice	Ea→Me	fwd 1	ORDER	2	0.069	3.47	0.9953		
Mars conjunction	S1a	Mercury (R8)	Me→Ma	fwd 2	ORDER	3	0.125	3.71	0.9980		
Mars conjunction	S1d	Cara	Ce→Ma	direct	CINSTR	4	0.258	3.29	0.9956		
Mars conjunction	S1d	Alice	Ea→Me	fwd 1	CINSTR	2	0.069	3.47	0.9953		
Mars conjunction	S1d	Mercury (R8)	Me→Ma	fwd 2	CINSTR	3	0.125	3.71	0.9980		

E6. Quota, backlog and storage

Two quota classes. Each exchange may originate 66 backbone packets in any rolling 24 hours. A data packet that carries any record other than RECEIVED is *routine* and may be sent only while the exchange has made fewer than 44 originations in the last 24 hours. SYN packets, packets that carry only RECEIVED records, and application resubmissions may use all 66. So at least 22 originations a day always remain for recovery and receipts, however much new work is queued. The simulator enforces both limits (`transport.Quota, reserve=22`). When routine work is blocked, waiting receipts are taken out of the queue and sent in packets of their own.

Backlog experiment. `backlog.py` puts a queue of \$100 transfers into the outboxes at hour 0, which is the situation at the end of a long outage, and runs until everything is resolved (Table B33). Three hundred transfers to one peer fill 60 data packets of five, plus one SYN. Earth sends 44 originations at once and the remaining 17 when the first ones leave the 24-hour window, so all transfers are imported by hour 24.6 and resolved by hour 25.1. Mars uses 60 originations for the receipts, which it may send from its reserve. With random loss every run cleared, with a median of 26.8 hours. Traffic in both directions costs little extra, because receipts share packets with the reverse transfers. A backlog to all eight peers at once is limited by Earth’s own 44 routine originations a day: 2,400 transfers need about 490 packets, or 11 days, and they resolve by hour 248. The 1,200-transfer case shows the window: the first 512 transfers go out and the rest wait until receipts come back.

What “clears” means. A backlog has cleared when every transfer is credited at its destination and the source has its RECEIVED, so no unresolved state remains. Credit comes earlier than resolution by at most one route delay plus the receipt’s queueing. The claim in the design paper is therefore: *when no other traffic competes, 300 transfers to one peer are credited within about 25 hours and resolved within about 26 hours of the route reopening; competing traffic from the same exchange shares its 44 routine originations a day.*

Window and compaction. A source numbers its transfers to each destination consecutively, sends them in that order, and does not send transfer n while any transfer numbered $n - 512$ or lower is unresolved (equivalently, $n < u + 512$ for the lowest unresolved number u). Let u be the lowest unresolved number at the source. Every transfer below u has a RECEIVED, so it was credited and the destination’s watermark

Table B33: Backlog after an outage. All transfers are already in the outbox at hour 0. One peer: 300 transfers from Earth to Mars. Both ways: 300 each way between Earth and Mars. All peers: 300 from Earth to each of the 8 other exchanges. Over window: 1,200 from Earth to Mars. “All imported” is when the last one is credited at its destination and “All resolved” when the source holds every RECEIVED (hours, no loss). N is the number of transfers. Orig. and Peak are Earth’s originations in total and in its busiest 24 hours (limit 66, of which 44 routine). The lossy columns use 20 runs (5 for the last row): runs fully resolved within 40 days, median and maximum time to resolve, and (Peak any) the highest 24-hour origination count of any exchange in any run. Held counts transfers that waited for the window of 512.

Case	N	Orig.	Peak	Imported	Resolved	Cleared	Median	Max	Peak any	Held
One peer	300	61	44	24.6	25.1	20/20	26.8	30.0	52	0
Both ways	600	109	65	24.6	25.7	20/20	28.3	31.5	65	0
All peers	2,400	490	51	244.2	248.4	20/20	330.5	445.0	66	0
Over window	1,200	241	44	120.6	121.1	5/5	124.3	127.6	66	688

satisfies $W \geq u - 1$. The source has sent nothing at or above $u + 512$. The credited transfers above W therefore all lie in $[u, u + 511]$, so the destination never keeps more than 512 tombstones per source, about 23 KB, however long a gap stays open. A transfer held by the window has already been debited and is durable at the source; it is in transit and still owned by the client, exactly like a transfer waiting for an open link. The simulator implements the window (`protocol.XFER_WINDOW`); in the 1,200-transfer case 688 transfers were held and released in order.

Admission and reservation. An exchange has 4,096 export slots. A slot is in use while an export is unresolved or while it is *reserved* for a transfer the exchange has promised. Accepting a remote client’s contract instruction reserves one slot for that party’s payout home; the slot is freed if the contract never opens, and used by the payout otherwise. Accepting a remote client’s order with home delivery reserves one slot for its delivery. An order has at most one delivery in transit: shares filled while it travels are added to a pending amount, sent in one transfer when the RECEIVED returns, and the slot is kept while the order rests. Funding, withdrawal and close each need slots for their own transfers. An instruction is refused (“try later”) if the slots in use plus those it needs would exceed 3,276; local trading continues. A payout or delivery therefore always finds its slot already held. The live-order limits (32 per client, 256 per host) are enforced at order entry.

Returns of late top-ups. A transfer to a closed account is the one export the host cannot refuse in advance: the home may have sent it before it learned of the closure, and each of the eight other exchanges may have up to 512 transfers in flight, so their number is not bounded by the spare 820 slots. The host credits such a transfer to a *restricted* balance, still owned by the client and usable for nothing but its return, and records one return entry per closed account, asset and source; a later top-up for the same entry adds to it. A return is exported only while slots in use are below the hard cap of 4,096, oldest entry first; otherwise it waits. Because returns use only free slots and admission counts them, slots in use never exceed the cap, and the reserved slots of payouts and deliveries are never taken. The waiting entries are bounded by the number of closed accounts times assets times sources, each one balance row. Every waiting return keeps a funded owner: the money stays in the client’s restricted balance at the host until the return is exported, and is then in transit, still the client’s, until it is credited at home.

Synthetic capacity tests. saturation.py tests the ledger’s capacity rules, not the network scenario: it is not a compliant workload under the brief. It uses about 5,250 synthetic client identities instead of the six opening accounts, injects their instructions at the Mars ledger without the 12-packet direct quota or R6 retries (2,000 contract instructions alone would need at least 667 direct packets, against 72 a day for the six real clients), and places their cash at Mars directly instead of funding it. Backbone transport, its quotas and the transfer window are real. Within two hours Mars receives 2,000 contract instructions (1,000 contracts of 48 hours), 250 home-delivery buy orders filled three shares every 30 minutes, and 3,000 withdrawals. Table B34 compares the reservation rule with the earlier rule that counted only unresolved exports. Under the earlier rule every request was accepted and unresolved exports reached 4,986, above the cap for 123 hours: the payouts were owed but had no capacity. Under the reservation rule the withdrawals that did not fit were refused at the door, slots in use never exceeded 3,276, and every payout and delivery went out.

A second test fills Mars to the admission limit with 3,276 withdrawals in flight and then delivers 2,400 top-ups from all 8 other exchanges (300 each) to accounts already closed at Mars, three times the spare

Table B34: Synthetic capacity test at Mars (not a compliant network scenario): the reservation rule against the earlier admission rule.

Mars at its limits	Reservation	Unresolved only
Contract instructions accepted / refused	2,000 / 0	2,000 / 0
Delivery orders accepted / refused	250 / 0	250 / 0
Withdrawals accepted / refused	1,026 / 1,974	3,000 / 0
Payouts resolved	2,000 of 2,000	2,000 of 2,000
Shares delivered (transfers)	2,500 of 2,500 (500)	2,500 of 2,500 (500)
Peak unresolved exports	3,012	4,986
Peak slots in use (incl. reserved)	3,276	5,250
Hours above the 4,096 cap	0.0	122.5
Last RECEIVED (h)	560	800
Pending shares / reserved slots at end	0 / 0	0 / 0

slots. Slots in use peaked at 4,096 and never exceeded the cap; up to 1,419 returns waited at once. All 2,400 returns were resolved, the last at hour 745; every client’s \$1,000 was back at home, and no restricted balance remained.

Table B35 lists every bounded part of an exchange’s state.

Retention of request results. Withdrawal, funding, cancel and close requests carry a per-account request number and a creation time. The exchange keeps each result (request ID, outcome, terms digest and transfer ID, 57 bytes) for 60 days after the creation time, which covers the brief’s 30-day packet lifetime and its 30-day duplicate retention. A copy created more than 30 days ago is rejected on that ground alone, so a dropped result can never be executed twice. A closed account keeps a 52-byte closure record so that a late top-up is returned, at least until every return for it is resolved. Once the home exchange learns of the closure it accepts no more funding for that account, so the record is dropped when the host’s watermark for the home passes the last transfer the home sent before learning of it. A contract record becomes a tombstone once both payouts are resolved and is dropped 30 days after the opening deadline, when every copy of its instructions has expired. These retention rules are design rules: the simulator keeps every result and record for the whole run (at most 1,600 hours), so it does not exercise the 60-day expiry.

E7. Market data

Design. Every exchange holds the listing table and a board with the newest snapshot of each stock (R5). A client reads both by local access, at no quota cost; at the market itself the board is the live book. Snapshots travel only as the reply to a quote request: a client’s QUOTE instruction makes its exchange ask the market over the backbone if the board is older than the client accepts, at most once per stock per 6 hours. In S1 the client sends QUOTE together with its funding, so the request shares the funding transfer’s packet and the reply shares the RECEIVED’s. Packet counts, originations and completion times are identical to runs without market data, in S1 and in 60 lossy S1a, S1c, S1d and S1i runs, and the client reads the quote just before it orders (mark `quote_seen`).

Load test. `md_test.py` has a client at every settlement ask for every stock listed elsewhere (24 exchange–stock pairs) for 30 days, with random loss and maintenance, while a local trader at each market changes the book every 2 hours. Table B36 gives the waits and ages. With a request every 6 hours, which is the most an exchange sends, each market sent 14.4 replies a day and its busiest 24 hours used 23 originations including session setup. With one request a day it sent 4.0 and used at most 12. 47% of the 6-hourly requests were answered from the board at once. Waits are set by distance: a request and its reply each cross the backbone, and at Uranus and Neptune a new session’s handshake and losses of about a third per hop add a day or more. The quote never gates safety: an order’s limit price bounds what it pays.

Table B35: Bounded state at each exchange.

State	Limit	When the limit is reached	Note
Live orders	32 per client per host; 256 per host	Order refused	Orders expire within 30 days
Export slots (unresolved exports plus reserved payouts and deliveries)	4,096 per host	An instruction needing slots is refused if slots in use would exceed 3,276 (80\%)	Payouts and deliveries hold their slot from acceptance; returns of late top-ups use only free slots and otherwise wait
Late top-ups awaiting return	One restricted balance per closed account, asset and source	Waits, owned by the client, until a slot is free	Later top-ups merge into it; never more slots than the cap
Transfers in flight to one destination	Window of 512	Transfer n waits (debited, durable) while any transfer numbered n – 512 or lower is unresolved	Sent in number order; bounds the destination’s tombstones
Transfer tombstones above the watermark	At most 512 per source	Cannot be exceeded	45 bytes each: ID 12, state 1, terms digest 32
Request results (withdrawal, funding, cancel, close)	Kept 60 days after the request’s creation time	A copy older than 30 days is rejected as expired	At most 64 requests per account per host per day
Account closure record	52 bytes until the host’s watermark for the home passes the home’s last transfer sent before it learned of the closure, and every return is resolved	—	Diverts late top-ups; the home sends none after (not simulated)
Contract records	Full record until both payouts are resolved, then a tombstone until 30 days after the opening deadline	—	Every instruction copy has expired by then (not simulated)

Table B36: Quote requests by clients at each exchange, pooled over the stocks listed elsewhere, in 10 lossy 30-day runs per demand level. Wait: hours from the QUOTE instruction to the quote (0 when the board was already fresh enough). Age: hours since the market took the snapshot, when the client received it. Daily: one request per stock per day, accepting quotes up to 24 hours old. 6 h: one every 6 hours, accepting quotes up to 6 hours old.

Exchange	Daily wait 95%	Daily age median	6 h wait 95%	6 h wait max	6 h age median	6 h age 95%
Mercury	3.3	19.7	3.2	8.0	2.7	5.3
Venus	2.8	16.9	2.8	7.8	2.3	5.4
Earth	2.7	19.2	2.6	5.8	2.4	5.5
Mars	2.6	20.7	2.6	4.3	2.3	5.5
Ceres	2.6	19.1	2.5	5.5	2.4	5.5
Jupiter	4.4	13.1	3.6	9.8	2.8	5.4
Saturn	5.7	12.7	5.7	29.3	3.2	4.7
Uranus	18.0	10.4	19.7	51.2	3.4	13.8
Neptune	43.4	12.9	49.4	84.5	5.1	30.4